

Segmentation of cave paintings

Project Proposal

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1. Motivation and definition of the subject

Paleolithic cave paintings represent a major component of prehistoric cultural heritage and are of great interest to archaeologists and art historians. Their study often requires isolating painted motifs from the surrounding rock surface, a task that is traditionally performed manually. This manual segmentation is time-consuming, subjective, and difficult to reproduce, especially when dealing with large image collections or comparative studies across sites.

The problem is further complicated by the poor visibility of many paintings. Due to aging, erosion, mineral deposits, and lighting conditions, pigments may appear faint or partially invisible to the naked eye. Image enhancement techniques such as decorrelation stretch [5] are now commonly used to reveal hidden or degraded pigments, but they remain exploratory tools and do not provide automated segmentation of the painted regions [2, 4].

From a computer vision perspective, segmentation of cave art has mainly been studied in the context of *petroglyphs*. Existing works focus on engraved symbols using 3D surface data, gradient-based methods, or deep learning approaches [1, 11]. In contrast, the automatic segmentation of Paleolithic cave paintings from 2D images has received little to no attention in the literature. The visual characteristics of paintings—diffuse pigments, low contrast, and ambiguous boundaries—make this problem distinct from petroglyph segmentation and unsuitable for direct reuse of existing methods.

Developing automatic segmentation methods for Paleolithic cave paintings would therefore address a clear gap in the literature and could enable large-scale quantitative studies of prehistoric art, contributing to both cultural heritage preservation and archaeological research.

2. Methodology

The objective of this project is to implement and evaluate segmentation methods for Paleolithic cave paintings. It will focus on classical image processing techniques rather

than deep learning approaches.

2.1. Overall approach

This work will be structured around at least two main families of approaches: histogram-based methods and contour-based methods.

All methods will follow a common processing pipeline:

- image preprocessing,
- segmentation using the selected method,
- post-processing to refine segmentation masks,
- evaluation.

2.2. Preprocessing

Cave painting images exhibit strong variability in illumination, color degradation, surface texture, and acquisition conditions. To mitigate these effects, several preprocessing steps will be investigated:

- color space conversion (RGB to grayscale, HSV, or Lab),
- contrast enhancement using histogram equalization and decorrelation stretching [5],
- noise reduction using Gaussian or median filtering.

These steps aim to enhance the visual contrast between painted regions and the rock background.

2.3. Histogram-based segmentation methods

Cave paintings often differ from the background in terms of color or intensity distribution, histogram-based methods can therefore be used.

Global thresholding methods such as Otsu’s algorithm [9] can be used to automatically determine an optimal threshold separating the paintings from the background. In addition, adaptive thresholding methods can be considered to account for spatial illumination variations commonly observed in caves.

2.4. Clustering-based segmentation methods

For the same reasons given in Sec. 2.3, clustering-based segmentation methods are also relevant for the segmentation of cave paintings. In particular, the mean-shift method

will be used, looking for clusters of pixels in a chosen color space in order to identify different "classes": cave wall, charcoal paintings, red-pigmented paintings...

2.5. Contour-based segmentation methods

Even though cave paintings often exhibit ill-defined boundaries relative to the surrounding rock surface, contour-based methods can be tested. Classic edge detection techniques such as Sobel or Canny filters [3] would be hard to use in this context due to the diffuse nature of the edges. However we plan to use the edges as input for region-based methods, including active contours (snakes) [6]. Such approaches are particularly well suited to segment irregular and fragmented shapes, which are common in Paleolithic art.

Morphological operations (closing, filling, erosion) can be applied as post-processing steps to transform detected contours into segmentation masks.

3. Evaluation

3.1. Database

As described in Sec. 1, automatic segmentation of Paleolithic cave paintings has, to the best of our knowledge, not been addressed in previous works.

Existing datasets in the field of cave art imaging mainly focus on *petroglyphs*, rather than painted motifs. Among the most well-known datasets is the *3D-Pitoti* dataset [10], which contains 26 high-resolution 3D surface reconstructions of rocks with annotated petroglyphs. Other resources such as the RockArt Database or the Swedish Rock Art Research Archives (SHFA)¹ also provide material for the study of petroglyphs. However, these datasets are not suitable for this project, as they do not include cave paintings.

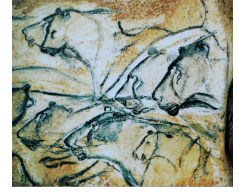
This project specifically focuses on the segmentation of Paleolithic cave paintings, dating from approximately 40,000 to 12,000 BP. This period corresponds to the emergence of cave art in Europe with symbolic representations of animals, hand stencils, and abstract signs. These paintings are of major archaeological importance, as they constitute some of the earliest known examples of human artistic expression. [7]

To the best of our knowledge, no annotated image dataset currently exists for the segmentation of Paleolithic cave paintings. Consequently, a dataset has been created for this project.

The study will focus on at least four major decorated caves located in France and Spain: Altamira Cave (Spain, 35,000 to 11,000 BP), Chauvet Cave (France, around 30,000 BP), Cosquer Cave (France, 33,000 to 19,000 BP) and Lascaux Cave (France, 21,000 to 17,000 BP). These caves played a crucial role in the establishment of the



(a) Altamira Cave painting.



(b) Panel of the lions - Chauvet Cave.

Figure 1. Examples of samples from the created dataset.

chronological framework of European prehistoric art and are considered reference sites in Paleolithic archaeology.

The dataset is constructed using web scraping to collect publicly available images of cave paintings. Special care was taken to ensure that all collected images are free of use or distributed under permissive licenses. At the time of writing, a total of 49 images have been collected. Examples of images from the dataset are shown in Fig. 1.

3.2. Metrics

The goal of the evaluation is to assess how accurately the proposed algorithms segment cave paintings from the background rock surface.

In a first stage, segmentation quality will be assessed qualitatively by visual inspection to quickly assess the quality of the results. However, this approach does not allow for a rigorous comparison between different algorithms.

To objectively evaluate the results, it would be better to use metrics based on the comparison between the predicted segmentation and the ground truth. This requires an annotated dataset, created from scratch using an annotation tool to produce ground truth masks. It would however take a very long time to annotate the whole dataset. A focus will be made on a few specific images that will constitute a testing database.

The evaluation will rely on metrics commonly used in the image segmentation literature: the Dice Similarity Coefficient (DSC) [1, 10, 11] and the mean Intersection over Union (mIoU) [1, 8].

While closely related to the DSC, the mIoU penalizes over-segmentation more strongly and is therefore complementary. Using both metrics provides a more comprehensive evaluation of segmentation quality.

3.3. Experiments

To ensure a fair and reproducible evaluation, a strict experimental protocol will be followed. The dataset will be split into a training set (80%) and a test set (20%), with no overlap between the two. All algorithms will be trained exclusively on the training set and evaluated on the test set using the metrics defined in Sec. 3.2.

¹<https://www.shfa.se>

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